

Quantitative Spatial Models for Transport Appraisal

Optional capstone add-on: Berlin's U5 connection

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Draft skeleton

Policy question

How should we appraise a transport investment when households, firms, wages, rents, employment, and location choices respond?

- ▶ Conventional appraisal emphasizes time savings and user benefits.
- ▶ A quantitative spatial model adds endogenous sorting and general-equilibrium adjustment.
- ▶ The aim is not to replace engineering appraisal, but to make wider spatial mechanisms explicit and internally consistent.

A quantitative transport-appraisal workflow

1. Define two comparable transport networks.
2. Compute bilateral travel-time matrices.
3. Validate identifiers, units, orientation, and routing.
4. Map travel times into commuting costs.
5. Quantify or invert a baseline spatial equilibrium.
6. Solve the policy equilibrium.
7. Report direct, equilibrium, distributional, and welfare effects.

Running example: Berlin's central U5 connection

- ▶ The complete network connects Alexanderplatz and Hauptbahnhof by U-Bahn.
- ▶ The comparison network removes the entire U-Bahn connection between these endpoints.
- ▶ Removing the full corridor creates a sufficiently informative accessibility shock for teaching and simulation.
- ▶ All other network inputs and routing settings remain fixed.

Interpretation

This is a deliberately stylised network counterfactual, not an ex-post causal estimate of the realised U5 extension.

Network construction

Complete network

- ▶ Current Berlin line and station layers
- ▶ Alexanderplatz–Hauptbahnhof connection retained

Comparison network

- ▶ Copy the line layer in QGIS
- ▶ Delete all corridor segments between the endpoints
- ▶ Preserve intersecting services
- ▶ Treat observed stations consistently

From GIS to bilateral travel times

- ▶ TTMATRIX builds routable networks from line and station shapefiles.
- ▶ Use identical blocks, speeds, transfers, access assumptions, and numerical settings in both scenarios.
- ▶ Export matrices using stable block identifiers.
- ▶ Record network length, station count, matrix dimensions, missing routes, and summary travel times.

Reproducibility rule: The scenario definition should be the only difference between the two runs.

Figure 1: policy geometry

Figure placeholder

Map the complete network in a light colour, highlight the Alexanderplatz–Hauptbahnhof corridor, and mark both endpoints. Add an inset showing the removed segments.

Figure 2: direct travel-time effects

Figure placeholder

Map an accessibility statistic such as the employment-weighted mean change in travel time by origin. Use a transparent scale and report units.

Matrix validation before economics

- ▶ Same dimension and exact identifier ordering.
- ▶ Minutes, not seconds; documented diagonal convention.
- ▶ Residence-by-workplace orientation consistent with ARSW.
- ▶ No unexplained missing or infinite OD pairs.
- ▶ For affected OD pairs, complete-network travel times should weakly improve relative to the reduced network.

Bridge to the ARSW model

Let t_{ij} denote travel time from residence i to workplace j .

1. Reorder both TTMATRIX outputs to the ARSW block index.
2. Convert t_{ij} into the commuting-cost object used by the model.
3. Treat the reduced network as the baseline transport system.
4. Recover or re-invert baseline fundamentals under that system.
5. Introduce the complete network while holding the chosen fundamentals and parameters fixed.

Equilibrium outcomes

- ▶ Residence and workplace employment
- ▶ Wages and expected worker income
- ▶ Floor-space prices and development density
- ▶ Commuting flows and average commuting costs
- ▶ Productivity and amenity feedbacks under endogenous agglomeration
- ▶ Welfare under closed- and open-city closures

Figure 3: employment responses

Figure placeholder

Two-panel map: percentage change in workplace employment and residence employment. Use common spatial support and distinguish levels from percentage changes.

Figure 4: prices, wages, and welfare

Figure placeholder

Candidate panels: floor-space prices, wages, and a welfare or expected-income measure. Add an aggregate summary table alongside the maps.

Mechanism decomposition

1. Direct change in travel times, before relocation.
2. Equilibrium with exogenous productivity and amenities.
3. Equilibrium with endogenous productivity and amenity spillovers.
4. Closed-city versus open-city closure.

This decomposition separates the network shock from model amplification and migration assumptions.

Robustness and limitations

- ▶ Alternative access, transfer, waiting-time, and mode assumptions.
- ▶ Alternative treatment of stations along the removed corridor.
- ▶ Sensitivity to commuting elasticity and agglomeration parameters.
- ▶ Comparison with the toolkit's original travel-time matrix.
- ▶ Travel time is not the full generalised cost of travel.
- ▶ Construction costs, capacity, reliability, and transition dynamics remain outside the current exercise.

Planned MATLAB outputs

Output	Transformation	Proposed figure
Travel time	minutes / weighted mean	accessibility map
Workplace employment	$100\Delta \log L^W$	spatial map
Residence employment	$100\Delta \log L^R$	spatial map
Wages	$100\Delta \log w$	spatial map
Floor-space prices	$100\Delta \log Q$	spatial map
Welfare	closure-specific	table and distribution

To be populated after the matrices have been validated and the baseline inversion has been agreed.

Capstone task

1. Produce and audit the two network scenarios.
2. Compute and validate both travel-time matrices.
3. Align them to the ARSW data.
4. Solve the complete-network counterfactual.
5. Explain mechanisms, distributional patterns, and limitations.

Deliverables: reproducible network files, audit table, aligned matrices, code, four core figures, and a concise policy interpretation.

Next development step

Joint work after the summer-school materials are stable

Run the validated matrices through the MATLAB workflow, agree the baseline and closure, create publication-quality maps and summary tables, and replace the placeholders in this deck.